

Unique completion as a quantization condition

Draft. Not submitted. Not a deposited preprint.

A process function has exactly one fixed point for every local intervention.

No fixed point is the grandfather case. Two is the bootstrap.

Baumeler, Costa, Ralph, Wolf and Zych extend every such map to a bijection

and leave the quantum step open. This file is the longer draft.

Claim A is checked. Claim B is written. Claim C is open.

Lean: [github.com/drewarrowood/process-function-unitary-lift](https://github.com/drewarrowood/process-function-unitary-lift/blob/main/lean/Full.lean)

[blob/main/lean/Full.lean](https://github.com/drewarrowood/process-function-unitary-lift/blob/main/lean/Full.lean)

1. The demand

Dummett set the dictionary objection aside in 1954 and again in 1964.

A stipulation about the word cause is not a discovery about the world.

Black: an agent who has seen the earlier event can try to prevent the later one.

Lewis splits the modal and does not select the histories.

Horwich asks a theory to say what happens to the intervention.

Faye, Stanford Encyclopedia, revision 28 October 2025, still treats the consistency paradoxes as unresolved.

Only consistent loops occur can mean a discard, a filter, or a definition of the admitted set as the maps with unique fixed points.

The third reading can be failed. The identity loop fails it. The swap fails it.

Unique completion is not an open choice.

2. Spacetime

Godel 1949 has closed timelike curves. The field equation does not forbid them.

Morris, Thorne and Yurtsever convert a traversable wormhole into a time machine if the throat is held by exotic stress-energy. That is a construction.

Friedman and collaborators keep only self-consistent solutions.

Echeverria, Klinkhammer and Thorne find one continuation, two, or none.

Carlini and collaborators recover the same underdetermination from the action.

Earman: a condition imposed after the dynamics is not a dynamics.

Hawking's chronology protection is a different question.

3. Three quantum replies

Deutsch asks for a fixed point of the partial trace. Existence, not uniqueness.

Aaronson and Watrous place the power at PSPACE.

Lloyd and collaborators replace the fixed point by post-selection.

That is a simulation. The probability of success is not one.

Oreshkov, Costa and Brukner define a process with no global order.

The quantum switch does not violate a causal inequality. Lugano does.

Price, Leifer and Pusey, and Maudlin are adjacent, not answered.

4. Equations

(1) $o = f(w(o))$.

The output returns itself. No solution is the grandfather case.

(2) $w(a, b, c) = (\text{not } b \text{ and } c, \text{not } c \text{ and } a, \text{not } a \text{ and } b)$.

Each input depends on the other two outputs. Two-to-one.

(3) (o, e) maps to $(w(o) \text{ XOR } e, o)$.

(4) (i, s) maps to $(s, i \text{ XOR } w(s))$.

On bits exclusive or cancels. Lean checked both compositions.

(5) some party disagrees in the output and agrees in w . Checked.

(6) $M \circ M\text{-dagger} = I$. Written, not kernel-checked.

5. The contraction

Put a local unitary at each party. The amplitude is a product, because each unitary sees only its own symbol.

Change variables by exclusive or with the process output.

The shift is $w(s')$ XOR $w(s)$.

The gram entry is the product of row overlaps against that shift.

On the diagonal the shift vanishes. Off it, the lemma supplies a party with shift zero and distinct rows, so that factor is zero.

The integer rows $(1, 1)$ and $(1, -1)$ have dot product 0.

A normalised Hadamard is not in the file.

6. Embeddability

A channel embeds as free propagation. The twelve binary two-party process functions are channels. No wormhole is required.

Lugano violates a causal inequality, so no definite order embeds it.

Post-selection is a simulation. The bijection is reversibility.

The billiard has the wrong multiplicity. No other local Hamiltonian is known.

The absence is not a proof that none exists.

7. Not claimed, and references

Not a choice for the agent. Not a solution of Leifer and Pusey.

Not a kernel check of (6). Not a refereed theorem.

Dummett 1954, 1964. Black 1956. Lewis 1976. Horwich 1987.

Faye, SEP, 28 October 2025. Godel 1949. Morris, Thorne, Yurtsever 1988.

Friedman et al. 1990. Echeverria, Klinkhammer, Thorne 1991.

Earman 1995. Hawking 1992. Deutsch 1991. Aaronson and Watrous 2009.

Lloyd et al. 2011. Oreshkov, Costa, Brukner 2012.

Baumeler and Wolf 2016. Baumeler, Costa, Ralph, Wolf, Zych 2019.

Araujo, Feix, Navascues, Brukner 2017. Leifer and Pusey 2017.

Dourdent et al. arXiv:2609.39735. Daher Ahmed and Kunjwal arXiv:2610.00579.

Stinespring 1955. Lean file: lean/Full.lean